Def. Let G be a Group, $x, y \in G$, $A, B \subseteq G$.

1) $[x,y] \overset{def}{=} x^{-1}y^{-1}xy$, commutator of x and y.

2). $[A,B] \overset{def}{=} \langle [a,b] \mid a \in A, b \in B \rangle$

3). $G' = \langle [x,y] \mid x, y \in G \rangle$.

Prop. Let G be a Group, $x, y \in G$, $H \leq G$.

1). $xy = yx[x,y]$. In particular, x, y are commute iff $[x,y]=1$.

2) $H \trianglelefteq G$ iff $[H,G] \leq H$.

3). $\forall \sigma \in Aut(G)$, $\sigma[x,y] = [\sigma(x), \sigma(y)]$

4). G' char G

5). G/G' is the largest Abelian Quotient of G, that is,

$H \trianglelefteq G$ and G/H Abelian iff $G' \leq H$.

6. If $\varphi: G \to A$ is Homomorphism into an Abelian Group A, then $G' \leq \ker \varphi$.

Proof. 1) is trivial, let's show the 2).

$H \trianglelefteq G \iff \forall g \in G, h \in H, ghg^{-1} \in H$.

$\iff \forall g \in G, h \in H, ghg^{-1}h^{-1} \in H$

$\iff \forall g \in G, h \in H, [h,g] \in H$

$\iff [H,G] \leq H$.

3). $\sigma[x,y] = \sigma(x^{-1}y^{-1}xy) = \sigma(x)^{-1}\sigma(y)^{-1}\sigma(x)\sigma(y) = [\sigma(x), \sigma(y)]$

4). Directly, since 3).

5) First, show Normality. By 2), $G' \trianglelefteq G$ iff $[G', G] \leq G'$. The right term is trivial:

$x \in [G', G] \iff x = [x_1, y_1] \cdots [x_n, y_n]$ for some $x_i \in G'$, $y_i \in G$.

Clearly $x_i \in G$, $x \in G'$.

Second, G/G' is Abelian, because: let $g_1G', g_2G' \in G/G'$ be given.

Then, $g_1G' \cdot g_2G' = g_1g_2G' = g_2g_1[g_1, g_2]G' = g_2g_1G' = g_2G' \cdot g_1G'$.

Finally, to show the largest: let $H \trianglelefteq G$ and G/H Abelian. Then, for any $x, y \in G$,

$1 \cdot H = (x^{-1}H)(y^{-1}H)(xH)(yH) = [x,y]H \iff [x,y] \in H$. Thus $G' \leq H$.

Conversely, suppose $G' \leq H$. Since G/G' is Abelian, Every subgroup of G/G' is Normal.

That is, $G/H \trianglelefteq G/G'$, this implies $H \trianglelefteq G'$.

Recall. for $H, K \leq G$, if each are finite,

$$|HK| = \frac{|H||K|}{|H \cap K|}.$$

Note: $H \to gH : h \mapsto gh$ bijective.
$a \sim b \iff ab^{-1} \in H$ is Equi. rel.

proof. $HK = \bigcup_{h \in H} hK$ clearly. And, for $h_1, h_2 \in H$,

$$h_1K = h_2K \iff h_1h_2^{-1} \in H \cap K \iff h_1(H \cap K) = h_2(H \cap K).$$

Now, $$|HK| = \left| \bigcup_{h \in H} hK \right| = [H : H \cap K] \cdot |K| = \frac{|H|}{|H \cap K|} \cdot |K|.$$

Thm. Let $K, H \leq G$, $x \in HK$.

$$|\{(h,k) \mid h \in H, k \in K, hk = x\}| = |H \cap K|. \qquad |HK| = \frac{|H||K|}{|H \cap K|}$$

Proof. Let $S := H \times K$. Then, clearly $|S| = |H||K|$.

Define a relation on S as:

$(h_1, k_1) \sim (h_2, k_2) \iff h_1k_1 = h_2k_2$.

Then, clearly $\sim$ is Equi. rel. on S. Now, consider a Partition:

$$S/\sim = \{[(h,k)] \mid h \in H, k \in K\}.$$

And, for any classes $[(h,k)]$, there exists a bijection from $[(1,1)]$ into $[(h,k)]$:

$$\varphi : [(1,1)] \to [(h,k)] : (a, a^{-1}) \mapsto (ha, a^{-1}k).$$

well-defined: $[(1,1)] = \{(h,k) \mid h \in H, k \in K, hk = 1\}$

$= \{(a, a^{-1}) \mid a \in H \cap K\}$. And, $haa^{-1}k = hk$.

injective: $(ha, a^{-1}k) = (hb, b^{-1}k) \Rightarrow a = b$.

surjective: Let $(h_1, k_1) \in H \times K$ be a tuple s.t. $h_1k_1 = hk$.

Then, $h^{-1}h_1 = kk_1^{-1} \in H \cap K$. Now,

$$\varphi(h^{-1}h_1, (h^{-1}h_1)^{-1}) = \varphi(h^{-1}h_1, k_1k^{-1}) = (hh^{-1}h_1, k_1k^{-1}k) = (h_1, k_1).$$

Consequently, for any class $[(h,k)] \in S/\sim$,

$$|S| = |H| \cdot |K| = |S/\sim| \cdot |[(h,k)]| = |H||K| \cdot |[(h,k)]| = \frac{|H||K|}{|H \cap K|} \cdot |[(h,k)]|$$

$$\Rightarrow |[(h,k)]| = |H \cap K|.$$

Recognition Theorem.

Suppose $H, K \trianglelefteq G$ are Normal Subgroup s.t $H \cap K = \{1\}$.

Then, $HK \cong H \times K$.

Proof. $\varphi : HK \to H \times K : hk \mapsto (h,k)$ is isomorphism, because:

1) $HK \leq G$.

$$HK = \bigcup_{h \in H} hK = \bigcup_{h \in H} Kh = KH$$

Now, for any $h_1k_1, h_2k_2 \in HK$, $h_1 \underbrace{k_1 k_2^{-1} h_2^{-1}}_{\in KH = HK} \in HK$.

2). $\forall h \in H, k \in K$, $hk = kh$.

Let $h \in H, k \in K$. Since Normality, $\begin{cases} k^{-1}hk \in H \Rightarrow h^{-1}k^{-1}hk \in H \\ h^{-1}k^{-1}h \in K \Rightarrow h^{-1}k^{-1}hk \in K \end{cases}$

That is, $[h,k] \in H \cap K = \{1\}$, h, k are commute.

3). Well-defined of HK.

By Theorem, each element of HK has the unique representation form

4). Injective and surjective.

Trivial.

5) Homomorphism.

$$\varphi(h_1k_1 \cdot h_2k_2) = \varphi(h_1h_2\, k_1k_2) = (h_1h_2, k_1k_2) = (h_1,k_1)\cdot(h_2,k_2) = \varphi(h_1,k_1)\cdot\varphi(h_2,k_2).$$

$\blacksquare$